$\boxed{1+1 \text{ dim.}}$ $\boxed{u_{tt} = u_{xx}, \quad u(0,x) = a(x), \quad u_t(0,x) = b(x)}$

$u(t,x) = \frac{1}{2}\left(a(x+t) + a(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} b(x')\,dx'$ とおくと,

$F'(x) = f(x)$ のとき

$\frac{d}{dx}\int_x^a f(x')\,dx' = \frac{d}{dx}\left[F(a) - F(x)\right] = -F'(x) = -f(x)$

$u(0,x) = a(x)$

$u_t(t,x) = \frac{1}{2}\left(a'(x+t) - a'(x-t)\right) + \frac{1}{2}\left(b(x+t) + b(x-t)\right)$

$u_t(0,x) = b(x)$

$u_{tt}(t,x) = \frac{1}{2}\left(a''(x+t) + a''(x-t)\right) + \frac{1}{2}\left(b'(x+t) - b'(x-t)\right)$

$u_x(t,x) = \frac{1}{2}\left(a'(x+t) + a'(x-t)\right) + \frac{1}{2}\left(b(x+t) - b(x-t)\right)$

$u_{xx}(t,x) = \frac{1}{2}\left(a''(x+t) + a''(x-t)\right) + \frac{1}{2}\left(b'(x+t) - b'(x-t)\right)$

等しい $u_{tt} = u_{xx}$.

$\boxed{u_{tt} = u_{xx} + f}$ $\hat{u} = \int_{\mathbb{R}} e^{-ipx} u\,dx, \quad \hat{u}_{xx} = -p^2\hat{u}$

$\hat{u}_{tt} = -p^2\hat{u} + \hat{f}, \quad \hat{u}(t,p) = \hat{a}(p), \quad \hat{u}_t(t,p) = \hat{b}(p)$

$\underline{\ddot{u} = -p^2 u + f(t) \text{ の解き方}}$

$u = c(t)e^{ipt} + d(t)e^{-ipt}$ とおく,

$\dot{u} = \dot{c}e^{ipt} + ipce^{ipt} + \dot{d}e^{-ipt} - ipde^{-ipt}$

$\ddot{u} = \ddot{c}e^{ipt} + 2ip\dot{c}e^{ipt} - p^2ce^{ipt} + \ddot{d}e^{-ipt} - 2ip\dot{d}e^{-ipt} - p^2de^{-ipt}$

$= -p^2u + \ddot{c}e^{ipt} + 2ip\dot{c}e^{ipt} + \ddot{d}e^{-ipt} - 2ip\dot{d}e^{-ipt}$

$\ddot{u} = -p^2u + f(t)$

$\begin{cases} \dot{u} = v \\ \dot{v} = -p^2u + f(t) \end{cases}$ $\frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}}^{=:A}\begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ f(t) \end{bmatrix}$

$U(t) := e^{t\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}} = \sum_{k=0}^{\infty}\frac{t^{2k}}{(2k)!}(-1)^k p^{2k}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \sum_{k=0}^{\infty}\frac{t^{2k+1}}{(2k+1)!}(-1)^k p^{2k}\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}$

$= \cos(pt)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{p}\sin(pt)\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}$

$w = U(t)c(t)$ とおくと $\dot{w} = Aw + U(t)\dot{c}(t)$

$U(t)\dot{c}(t) = g(t)$

$\dot{c}(t) = U(-t)g(t), \quad c(t) = c(0) + \int_0^t U(-t')g(t')\,dt'$

$c(0) = \begin{bmatrix} a(0) \\ b(0) \end{bmatrix}$, $g(t') = \begin{bmatrix} 0 \\ f(t') \end{bmatrix}$

$u(t) = a(0)\cos(pt) + \frac{b(0)}{p}\sin(pt) + \boxed{///}$

$$\boxed{u_{tt} = u_{xx} + f} \qquad \hat{u} = \int_{\mathbb{R}} e^{-ipx} u\, dx, \quad \hat{u}_{xx} = -p^2 \hat{u}$$

$$\hat{u}_{tt} = -p^2\hat{u} + \hat{f}, \quad \hat{u}(t,p) = \hat{a}(p), \quad \hat{u}_t(t,p) = \hat{b}(p)$$

$\underline{\ddot{u} = -p^2 u + f(t) \text{ の解き方}}$

$u = c(t)e^{ipt} + d(t)e^{-ipt}$ とおく,

$$\dot{u} = \dot{c}e^{ipt} + ipce^{ipt} + \dot{d}e^{-ipt} - ip\, de^{-ipt}$$

$$\ddot{u} = \ddot{c}e^{ipt} + 2ip\dot{c}e^{ipt} - p^2 ce^{ipt} + \ddot{d}e^{-ipt} - 2ip\dot{d}e^{-ipt} - p^2 de^{-ipt}$$

$$= -p^2 u + \ddot{c}e^{ipt} + 2ip\dot{c}e^{ipt} + \ddot{d}e^{-ipt} - 2ip\dot{d}e^{-ipt}$$

$$\ddot{u} = -p^2 u + f(t)$$

$$\begin{cases} \dot{u} = v \\ \dot{v} = -p^2 u + f(t) \end{cases} \qquad \frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}}^{=: A}\begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ f(t) \end{bmatrix}$$

$$U(t) := e^{t\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}} = \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!}(-1)^k p^{2k}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!}(-1)^k p^{2k}\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}$$

$$= \cos(pt)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{p}\sin(pt)\begin{bmatrix} 0 & 1 \\ -p^2 & 0 \end{bmatrix}$$

$$c(0) = \begin{bmatrix} a(0) \\ b(0) \end{bmatrix}$$

$w = U(t)c(t)$ とおくと $\quad \dot{w} = Aw + U(t)\dot{c}(t)$

$$U(t)\dot{c}(t) = g(t)$$

$$\dot{c}(t) = U(-t)g(t), \quad c(t) = c(0) + \int_0^t U(-t')\underbrace{g(t')}_{=\begin{bmatrix} 0 \\ f(t') \end{bmatrix}} dt'$$

$$u(t) = a\cos(pt) + \frac{b}{p}\sin(pt) + \left(U(t)\int_0^t U(-t')\begin{bmatrix} 0 \\ f(t') \end{bmatrix} dt' \text{ の第1成分}\right)$$

$$U(t)U(-t')\begin{bmatrix} 0 \\ f(t') \end{bmatrix} = U(t-t')\begin{bmatrix} 0 \\ f(t') \end{bmatrix} = \begin{bmatrix} \frac{1}{p}\sin(p(t-t')) \\ \cos(p(t-t')) \end{bmatrix} f(t')$$

$$\boxed{u(t) = a\cos(pt) + \frac{b}{p}\sin(pt) + \frac{1}{p}\int_0^t \sin(p(t-t'))f(t')\,dt'}$$

$$u(0) = a$$

$$\dot{u}(t) = -pa\sin(pt) + b\cos(pt) + \int_0^t \cos(p(t-t'))f(t')\,dt'$$

$$\dot{u}(0) = b$$

$$\ddot{u}(t) = \underbrace{-p^2 a\cos(pt) - pb\sin(pt) - p\int_0^t \sin(p(t-t'))f(t')\,dt'}_{= -p^2 u(t)} + f(t) = -p^2 u(t) + f(t)$$

OK

$$\boxed{\hat{u}_{tt} = -p^2\hat{u} + \hat{f}(t,p),\quad \hat{u}(t,p)=\hat{a}(p),\quad \hat{u}_t(t,p)=\hat{b}(p)}$$

$$\boxed{\hat{u}(t,p) = \hat{a}(p)\cos(pt) + \frac{1}{p}\hat{b}(p)\sin(pt) + \frac{1}{p}\int_0^t \sin(p(t-t'))\hat{f}(t',p)\,dt'}$$

$$u(t,x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\hat{u}(t,p)\,dp$$

$$\frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\cos(pt)\hat{a}(p)\,dp = \frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\cos(pt)\left(\int_{\mathbb{R}} e^{-ipy}a(y)\,dy\right)dp$$

$$= \int_{\mathbb{R}}\left(\frac{1}{2\pi}\int_{\mathbb{R}} e^{i(x-y)p}\cos(tp)\,dp\right)a(y)\,dy = \int_{\mathbb{R}}\left(\frac{1}{2\pi}\int_{\mathbb{R}} \frac{e^{i(x-y+t)p}+e^{i(x-y-t)p}}{2}\,dp\right)a(y)\,dy$$

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\left(\int_{\mathbb{R}} e^{-ipy}f(y)\,dy\right)dp = \int_{\mathbb{R}}\underbrace{\left(\frac{1}{2\pi}\int_{\mathbb{R}} e^{i(x-y)p}\,dp\right)}_{=\delta(x-y)=\delta(y-x)}f(y)\,dy$$

$$= \int_{\mathbb{R}}\frac{\delta(x-y+t)+\delta(x-y-t)}{2}a(y)\,dy = \frac{a(x+t)+a(x-t)}{2}$$

$$\int_{\mathbb{R}} e^{-ipy}\frac{1}{2}\chi_{[-t,t]}(y)\,dy = \frac{1}{2}\left[\frac{e^{-ipy}}{-ip}\right]_{y=-t}^{y=t} = \frac{e^{-itp}-e^{itp}}{-2ip} = \frac{\sin(tp)}{p}\qquad \boxed{t>0\text{と仮定}}$$

$$\frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\frac{\sin(tp)}{p}\,dp = \frac{1}{2}\chi_{[-t,t]}(x)$$

$$\frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\frac{\sin(tp)}{p}\hat{b}(p)\,dp = \int_{\mathbb{R}}\overbrace{\left(\frac{1}{2\pi}\int_{\mathbb{R}} e^{i(x-y)p}\frac{\sin(tp)}{p}\,dp\right)}^{=\frac{1}{2}\chi_{[-t,t]}(x-y)=\frac{1}{2}\chi_{[x-t,x+t]}(y)}b(y)\,dy = \frac{1}{2}\int_{x-t}^{x+t}b(y)\,dy$$

$$\frac{1}{2\pi}\int_{\mathbb{R}} e^{ipx}\frac{\sin(p(t-t'))}{p}\hat{f}(t',p)\,dp = \int_{\mathbb{R}}\overbrace{\left(\frac{1}{2\pi}\int_{\mathbb{R}} e^{i(x-y)p}\frac{\sin((t-t')p)}{p}\,dp\right)}^{=\frac{1}{2}\chi_{[-(t-t'),t-t']}(x-y)}f(t',y)\,dy$$

$$= \frac{1}{2}\int_{x-(t-t')}^{x+(t-t')}f(t',y)\,dy$$

$$\boxed{u(t,x) = \frac{1}{2}\left(a(x+t)+a(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t}b(y)\,dy + \frac{1}{2}\int_0^t dt'\int_{x-(t-t')}^{x+(t-t')}f(t',y)\,dy}$$

$$\boxed{u(t,x) = \frac{1}{2}\left(a(x+t) + a(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} b(y)\,dy + \frac{1}{2}\int_0^t dt' \int_{x-(t-t')}^{x+(t-t')} f(t',y)\,dy}$$

$$u(0,x) = a(x)$$

$$u_t(t,x) = \frac{1}{2}\left(a'(x+t) - a'(x-t)\right) + \frac{1}{2}\left(b(x+t) + b(x-t)\right) + \frac{1}{2}\int_0^t \left(f(t', x+(t-t')) + f(t', x-(t-t'))\right) dt'$$

$$u_t(0,x) = b(x)$$

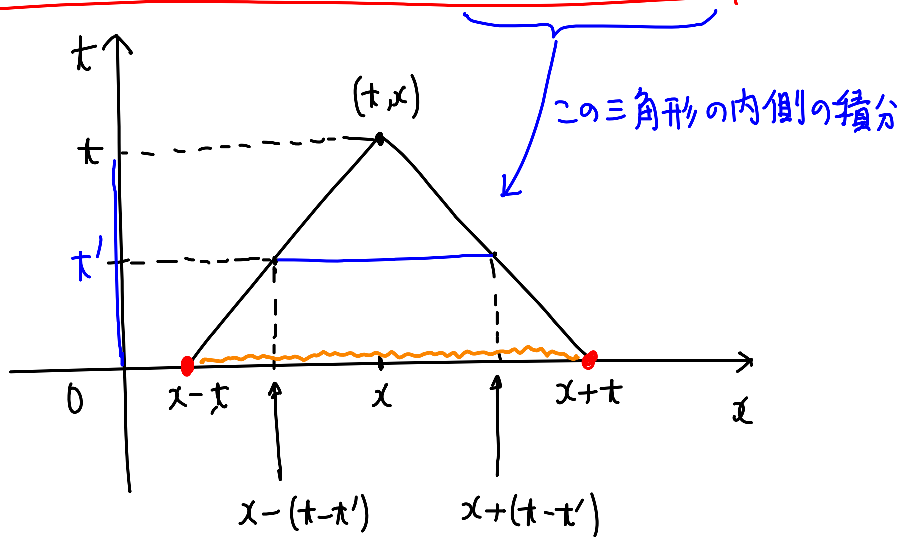
$$u_{tt}(t,x) = \frac{1}{2}\left(a''(x+t) + a''(x-t)\right) + \frac{1}{2}\left(b'(x+t) - b'(x-t)\right) + f(t,x) + \frac{1}{2}\int_0^t \left(f_x(t', x+(t-t')) - f_x(t', x-(t-t'))\right) dt'$$

$$u_x(t,x) = \frac{1}{2}\left(a'(x+t) + a'(x-t)\right) + \frac{1}{2}\left(b(x+t) - b(x-t)\right) + \frac{1}{2}\int_0^t \left(f(t', x+(t-t')) - f(t', x-(t-t'))\right) dt'$$

$$u_{xx}(t,x) = \frac{1}{2}\left(a''(x+t) - a'(x-t)\right) + \frac{1}{2}\left(b'(x+t) - b'(x-t)\right) + 0 + \frac{1}{2}\int_0^t \left(f_x(t', x+(t-t')) - f_x(t', x-(t-t'))\right) dt'$$

$$\therefore\ u_{tt}(t,x) = u_{xx}(t,x) + f(t,x) \qquad \boxed{\text{OK}}$$

$$\boxed{u(t,x) = \frac{1}{2}\left(a(x+t) + a(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} b(y)\,dy + \frac{1}{2}\int_0^t dt' \int_{x-(t-t')}^{x+(t-t')} f(t',y)\,dy}$$



1+1次元非斉次波動方程式の初期値問題

$$u_{tt}(t,x)=u_{xx}(t,x)+f(t,x),\qquad u(0,x)=a(x),\qquad u_t(0,x)=b(x)$$

の解の公式：

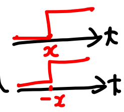
$$u(t,x)=\frac{1}{2}\bigl(a(x+t)+a(x-t)\bigr)+\frac{1}{2}\int_{x-t}^{x+t}b(y)\,dy+\frac{1}{2}\int_0^t ds\int_{x-(t-s)}^{x+(t-s)}f(s,y)\,dy.$$

Fourier変換に関する公式

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{ipx}\,dp\overset{①}{=}\delta(x),\quad \frac{1}{2\pi}\int_{-\infty}^{\infty}e^{ipx}\cos(pt)\,dp\overset{②}{=}\frac{1}{2}\bigl(\delta(x+t)+\delta(x-t)\bigr),\quad \frac{1}{2\pi}\int_{-\infty}^{\infty}e^{ipx}\frac{\sin(pt)}{p}\,dp\underset{t>0}{\overset{③}{=}}\frac{1}{2}\chi_{[-t,t]}(x).$$

① & $\cos(px)=\dfrac{e^{ipx}+e^{-ipx}}{2}$ $\Rightarrow$ ②

$t>0$のとき,

$\chi_{[-t,t]}(x)=\Bigl\{$ 

$(x>0)$, $(x<0)$ より

$\dfrac{\partial}{\partial t}\chi_{[-t,t]}(x)=\delta(x+t)+\delta(x-t)$ & ③ $\Rightarrow$ ②

非斉次線形常微分方程式の解の公式 ($p\neq0$のとき)

$\ddot{u}(t)=-p^2u(t)+f(t),\quad u(0)=a,\quad \dot{u}(0)=b$ の解は

$$u(t)\overset{(*)}{=}a\cos(pt)+b\frac{\sin(pt)}{p}+\int_0^t\frac{\sin(p(t-s))}{p}f(s)\,ds,$$

Proof (*)を仮定すると,

$$u(0)=a,\quad \dot{u}(t)=-pa\sin(pt)+pb\frac{\cos(pt)}{p}+p\int_0^t\frac{\cos(p(t-s))}{p}f(s)\,ds,\quad \dot{u}(0)=b.$$

$$\ddot{u}(t)=-p^2a\cos(pt)-p^2b\frac{\sin(pt)}{p}+f(t)-p^2\int_0^t\frac{\sin(p(t-s))}{p}f(s)\,ds=-p^2u(t)+f(t).\qquad\square$$

$u_{tt}(t,x)=u_{xx}(t,x)+f(t,x)$, $u(0,x)=a(x)$, $u_t(0,x)=b(x)$ の解法

$$\int_{-\infty}^{\infty}e^{-ipy}\frac{1}{2}\chi_{[-t,t]}(y)\,dy=\frac{1}{2}\int_{-t}^{t}e^{-ipy}\,dy=\left[\frac{e^{-ipy}}{-2ip}\right]_{y=-t}^{y=t}=\frac{\sin(pt)}{p}$$

$\Rightarrow$ ③

$\hat{g}(p):=\int_{-\infty}^{\infty}e^{-ipy}g(y)\,dy$ とおくと, $\widehat{g''}(p)=-p^2\hat{g}(p)$. ゆえに,

$$u_{tt}(t,x)=u_{xx}(t,x)+f(t,x),\quad u(0,x)=a(x),\quad u_t(0,x)=b(x)$$

$$\Leftrightarrow\quad \hat{u}_{tt}(t,p)=-p^2\hat{u}(t,p)+\hat{f}(t,p),\quad \hat{u}(0,p)=\hat{a}(p),\quad \hat{u}_t(0,p)=\hat{b}(p)$$

$$\overset{(*)}{\Leftrightarrow}\quad \hat{u}(t,p)=\hat{a}(p)\cos(pt)+\hat{b}(p)\frac{\sin(pt)}{p}+\int_0^t\frac{\sin(p(t-s))}{p}\hat{f}(s,p)\,ds$$

$$\overset{②,③}{\Leftrightarrow}\quad u(t,x)=\frac{1}{2}\int_{-\infty}^{\infty}\bigl(\delta(x-y+t)+\delta(x-y-t)\bigr)a(y)\,dy$$

$$+\frac{1}{2}\int_{-\infty}^{\infty}\chi_{[-t,t]}(x-y)\,b(y)\,dy$$

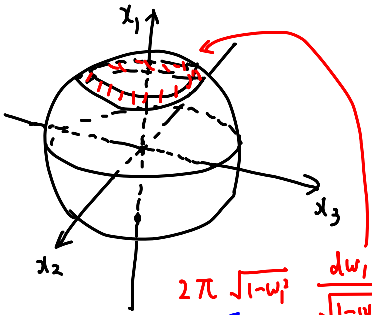
$$+\frac{1}{2}\int_0^t\left(\int_{-\infty}^{\infty}\chi_{[-(t-s),\,t-s]}(x-y)\,f(s,y)\,dy\right)ds$$

$$=\frac{1}{2}\bigl(a(x+t)+a(x-t)\bigr)+\frac{1}{2}\int_{x-t}^{x+t}b(y)\,dy+\frac{1}{2}\int_0^t ds\int_{x-(t-s)}^{x+(t+s)}f(s,y)\,dy.\qquad\square$$

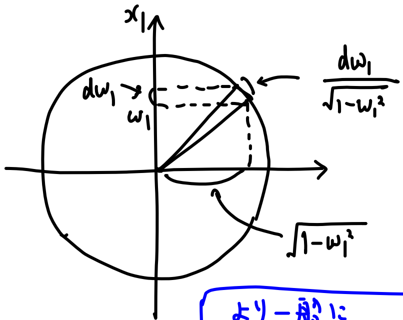
$$\boxed{\int_{S^2} e^{-ip\cdot r\omega}\,d\omega = \frac{4\pi\sin(r|p|)}{r|p|} \text{ を示そう.}}$$

$|\omega|=1$ 単位球面

$p=|p|e_1$, $r|p|=a$ の場合の $\displaystyle\int_{S^2} e^{-ia\omega_1}\,d\omega = \frac{4\pi\sin a}{a}$ を示せばよい.



S^{n-1} では
$A(S^{n-2})(1-\omega_1^2)^{\frac{n-3}{2}}\,d\omega_1$



$2\pi\sqrt{1-\omega_1^2}\,\dfrac{d\omega_1}{\sqrt{1-\omega_1^2}} = 2\pi\,d\omega_1$

$\boxed{(\sqrt{1-\omega_1^2})^{n-2}}$

一般に $\displaystyle\int_{S^2} f(\omega_1)\,d\omega = \int_{-1}^{1} f(\omega_1)\,2\pi\,d\omega_1 = 2\pi\int_{-1}^{1} f(\omega_1)\,d\omega_1$.

より一般に
$$\int_{S^{n-1}} f(\omega_1)\,d\omega_1 = A(S^{n-2})\int_{-1}^{1} f(\omega_1)(1-\omega_1^2)^{\frac{n-3}{2}}\,d\omega_1$$

$$\int_{S^2} e^{-ia\omega_1}\,d\omega_1 = \int_{-1}^{1} e^{-ia\omega_1}\,2\pi\,d\omega_1 = \pi\left[\frac{e^{-ia\omega_1}}{-ia}\right]_{\omega_1=-1}^{\omega_1=1} = 2\pi\,\frac{e^{-ia}-e^{ia}}{-ia} = 4\pi\,\frac{\sin a}{a} \quad \boxed{\text{OK}}$$

$$\int_{\mathbb{R}} f(x)\delta(ax)\,dx = \int_{\mathbb{R}} f\left(\frac{y}{a}\right)\delta(y)\,\frac{dy}{a} = \frac{f(0)}{a}$$

$\boxed{\text{公式}}$ $$\iiint_{\mathbb{R}^3} e^{-ip\cdot ry}\,\delta(|y|-1)\,dy_1dy_2dy_3 = 4\pi\,\frac{\sin(r|p|)}{r|p|}$$

$x=ry$ とおくと, $\delta\left(\dfrac{|x|-r}{r}\right) = r\,\delta(|x|-r)$, $dy_1dy_2dy_3 = \dfrac{1}{r^3}\,dx_1dx_2dx_3$ なので

$$\iiint_{\mathbb{R}^3} e^{-ip\cdot x}\,\delta(|x|-r)\,dx_1dx_2dx_3 = 4\pi r^2\,\frac{\sin(r|p|)}{r|p|}.$$

$\boxed{\text{公式}}$ 上の逆 Fourier 変換版:
$$\delta(|x|-r) = \frac{4\pi r^2}{(2\pi)^3}\iiint_{\mathbb{R}^3} e^{ip\cdot x}\,\frac{\sin(r|p|)}{r|p|}\,dp_1dp_2dp_3$$

ゆえに $t>0$ のとき,
$$\frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} dp\, e^{ip\cdot x}\,\frac{\sin(|p|t)}{|p|}\left(\int_{\mathbb{R}^3} e^{-ip\cdot y}\,b(y)\,dy\right) = \int_{\mathbb{R}^3} dy\,\underbrace{\left(\frac{t}{(2\pi)^3}\int_{\mathbb{R}^3} e^{ip(x-y)}\,\frac{\sin(|p|t)}{|p|}\,dp\right)}^{\frac{1}{4\pi t}\delta(|x-y|-t)}\,b(y)$$
$$= \frac{1}{4\pi t}\int_{\mathbb{R}^3} \delta(|y-x|-t)\,b(y)\,dy = \frac{1}{4\pi t}\int_{S^2} b(x+t\omega)\,t^2\,d\omega = \frac{t}{4\pi}\int_{S^2} b(x+t\omega)\,d\omega.$$

b を a でおきかえ, 両辺を t で微分すると,
$$\frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} dp\, e^{ipx}\cos(|p|t)\left(\int_{\mathbb{R}^3} e^{-ipy}\,a(y)\,dy\right) = \frac{\partial}{\partial t}\left(\frac{t}{4\pi}\int_{S^2} a(x+t\omega)\,d\omega\right)$$
$$= \frac{1}{4\pi}\int_{S^2} a(x+t\omega)\,d\omega + \frac{t}{4\pi}\int_{S^2} \omega\cdot a'(x+t\omega)\,d\omega \xmapsto{t=0} a(x). \quad \text{OK}$$

ゆえに, $\displaystyle\frac{\partial}{\partial t}\left(\frac{t}{4\pi}\int_{S^2} b(x+t\omega)\,d\omega\right) \xmapsto{t=0} b(x)$. OK

3次元の強制力付き波動方程式の解

$x=(x_1,x_2,x_3)$, $p=(p_1,p_2,p_3)$, $\Delta=\left(\frac{\partial}{\partial x_1}\right)^2+\left(\frac{\partial}{\partial x_2}\right)^2+\left(\frac{\partial}{\partial x_3}\right)^2$ とする.

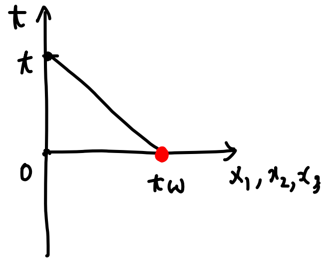
$$u_{tt}(t,x)=\Delta u(t,x)+f(t,x),\quad u(0,x)=a(x),\quad u_t(0,x)=b(x).$$

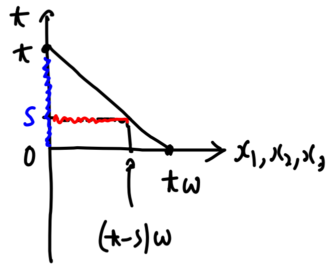
$$\hat{u}_{tt}(t,p)=-|p|^2\hat{u}(t,x)+\hat{f}(t,p),\quad \hat{u}(0,p)=\hat{a}(p),\quad \hat{u}_t(0,p)=\hat{b}(p).$$

$$\hat{u}(t,p)=\hat{a}(p)\cos(|p|t)+\hat{b}(p)\frac{\sin(|p|t)}{|p|}+\int_0^t\frac{\sin(|p|(t-s))}{|p|}\hat{f}(s,p)\,ds$$

$t>0$ とすると,

$$u(t,x)=\frac{\partial}{\partial t}\left(\frac{t}{4\pi}\int_{S^2}a(x+t\omega)\,d\omega\right)+\frac{t}{4\pi}\int_{S^2}b(x+t\omega)\,d\omega+\int_0^t\left(\frac{t-s}{4\pi}\int_{S^2}f(s,x+(t-s)\omega)\,d\omega\right)ds$$





1次元の場合

x_2, x_3 にすべてがよらないと仮定する.

$y=x_1+t\omega_1$, $dy=t\,d\omega_1$

$$\frac{t}{4\pi}\int_{S^2}b(x+t\omega)\,d\omega=\frac{t}{4\pi}\int_{-1}^{1}b(x_1+t\omega_1)\,2\pi\,d\omega_1=\frac{t}{4\pi}\int_{x_1-t}^{x_1+t}b(y)\,2\pi\frac{dy}{t}=\frac{1}{2}\int_{x_1-t}^{x_1+t}b(y)\,dy.$$

$$\frac{\partial}{\partial t}\left(\frac{t}{4\pi}\int_{S^2}a(x+t\omega)\,d\omega\right)=\frac{\partial}{\partial t}\left(\frac{1}{2}\int_{x_1-t}^{x_1+t}a(y)\,dy\right)=\frac{1}{2}\left(a(x_1+t)+a(x_1-t)\right).$$

$$\frac{t-s}{4\pi}\int_{S^2}f(s,x+(t-s)\omega)\,d\omega=\frac{1}{2}\int_{x_1-(t-s)}^{x_1+(t-s)}f(s,y)\,dy.$$

OK

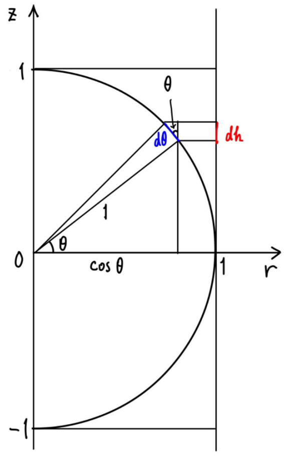
$$rS^{n-1}=\{(x_1,\dots,x_n)\in\mathbb{R}^n\mid x_1^2+\cdots+x_n^2=r^2\},\quad 1S^{n-1}=S^{n-1},\quad A(rS^{n-1})=A(S^{n-1})r^{n-1}$$

$$\frac{1}{A(rS^{n-1})}\int_{rS^{n-1}}f(\sigma)\,d\sigma=\frac{1}{A(S^{n-1})}\int_{S^{n-1}}f(r\omega)\,d\omega$$

$\sigma=r\omega$

$\boxed{n=3}$

$$\frac{1}{4\pi r^2}\int_{rS^2}f(\sigma)\,d\sigma=\frac{1}{4\pi}\int_{S^2}f(r\omega)\,d\omega.$$



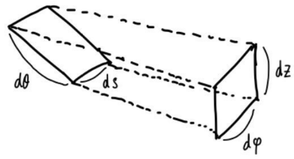
$\cos\theta : 1 = dh : d\theta$

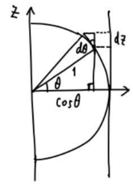
$\therefore\ 2\pi\cos\theta\, d\theta = 2\pi\, dh$

$\therefore$ (単位球の面積) = (円柱の側面積)

$= 2\pi\times 2 = 4\pi.$

三角函数の記号を使ったが，
$\cos\theta$ を a などと書けば，
三角函数の記号は必要ない
ことがわかる．
内容は直観的にわかりやすい．





$1 : \cos\theta = d\theta : dz = d\varphi : ds \quad \therefore\ d\theta\, ds = dz\, d\varphi$

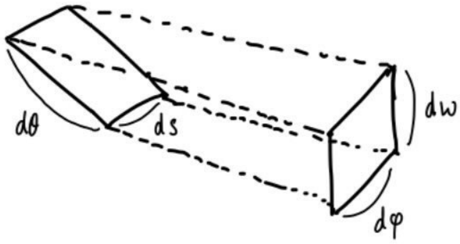
そして，$\displaystyle\int_{-1}^{1} dz \int_0^{2\pi} d\varphi = 4\pi$ ← 2次元単位球面 S^2 の面積

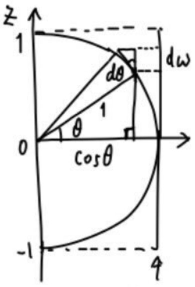
2次元球面の場合

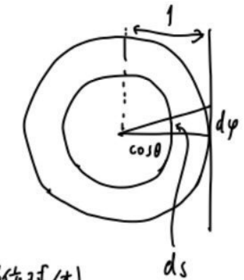
3次元球面では $1 : \cos\theta = d\theta : dz$ は同じだが，$1 : \cos^2\theta = d\varphi : ds$ となる．

$\therefore\ d\theta = \dfrac{dz}{\cos\theta}$，$ds = \cos^2\theta\, d\varphi$，$d\theta\, ds = \cos\theta\, dz\, d\varphi = \sqrt{1-z^2}\, dz\, d\varphi$

そして，$\displaystyle\int_{-1}^{1}\sqrt{1-z^2}\, dz \int_{S^2} d\varphi = \frac{\pi}{2}\times 4\pi = 2\pi^2.$







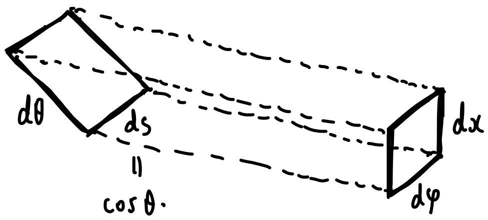
D^n = (n次元単位球体)

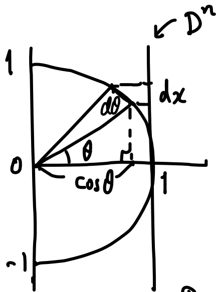
$S^{n-1} = \partial D^n$

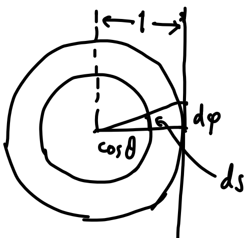
$1 : \cos\theta = d\theta : dw = d\varphi : ds$

$\therefore\quad ds\, d\theta = d\varphi\, dw$

$\therefore\quad$ ($\mathbb{R}^{n+2}$ 中の S^{n+1} の面積) = ($\mathbb{R}^{n+2}$ 中の $S^1\times D^n$ の面積) $= 2\pi$ ($\mathbb{R}^n$ 中の D^n の体積).







$dx = \cos\theta\, d\theta$ $\qquad$ $ds = \cos\theta\cdot d\varphi$

$1 : \cos\theta = d\theta : dx = d\varphi : ds \quad \therefore\ d\theta\, ds = \underbrace{dx\, d\varphi}_{\cos\theta\, d\theta\, d\varphi\ =:\ dw}$

$\sqrt{x_1^2+\cdots+x_n^2} = \sin\theta$，$\cos\theta = \sqrt{x_{n+1}^2 + x_{n+2}^2}$

$f(x_1,\dots,x_n,\cos\theta\cdot\cos\varphi,\cos\theta\cdot\sin\varphi)$ が φ によらないとき，$f\, dx = d\alpha$ とすると，

$$\int_{S^{n+1}} f\, dw = \int_0^{2\pi} d\varphi \int_{D^n} f\, dx = 2\pi\int_{D^n} f\, dx \overset{\sim}{=} 2\pi\int_{S^{n-1}} \alpha$$

$$\int_{-\infty}^{\infty} e^{ipx}\frac{\sin(ap)}{p}dp = \int_{-\infty}^{\infty} e^{ipx}\frac{e^{iap}-e^{-iap}}{2ip}dp = \int_{-\infty}^{\infty}\frac{e^{i(x+a)p}-e^{i(x-a)p}}{2ip}dp$$

$a>0$とする。

$$= \int_{-\infty+\varepsilon i}^{\infty+\varepsilon i}\frac{e^{i(x+a)p}-e^{i(x-a)p}}{2ip}dp$$

$$\begin{cases} -a<x<a \\ \Leftrightarrow 0<x+a<2a \\ \Leftrightarrow -2a<x-a<0 \end{cases}$$

$$= \int_{-\infty+\varepsilon i}^{\infty+\varepsilon i}\frac{e^{i(x+a)p}}{2ip}dp - \int_{-\infty+\varepsilon i}^{\infty+\varepsilon i}\frac{e^{i(x-a)p}}{2ip}dp$$

$$\begin{cases} x<-a \text{ or } a<x \\ \Leftrightarrow x+a<0 \text{ or } 0<x-a \end{cases}$$

$x+a<0$のとき、$x-a<0$となり、$p=\frac{z}{-(x+a)}$, $p=\frac{z}{-(x-a)}$ とおくと、

$$= \int_{-\infty+\varepsilon' i}^{\infty+\varepsilon' i}\frac{e^{-iz}}{2iz}dz - \int_{-\infty+\varepsilon'' i}^{\infty+\varepsilon'' i}\frac{e^{-iz}}{2iz}dz = 0$$

$x-a>0$のとき、$x+a>0$となり、$p=\frac{z}{x+a}$, $p=\frac{z}{x-a}$ とおくと、

$$= \int_{-\infty+\varepsilon' i}^{\infty+\varepsilon' i}\frac{e^{iz}}{2iz}dz - \int_{-\infty+\varepsilon'' i}^{\infty+\varepsilon'' i}\frac{e^{iz}}{2iz}dz = 0$$

$x+a>0$ かつ $x-a<0$ のとき、$p=\frac{z}{x+a}$、$p=\frac{z}{-(x-a)}$ とおくと

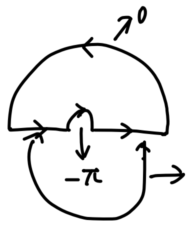
$$= \int_{-\infty+\varepsilon' i}^{\infty+\varepsilon' i}\frac{e^{iz}}{2iz}dz - \int_{-\infty+\varepsilon'' i}^{\infty+\varepsilon'' i}\frac{e^{-iz}}{2iz}dz$$

$$\int_0^{\infty}\frac{\sin x}{x}dx = \frac{\pi}{2}$$

$$= \int_{-\infty+\varepsilon' i}^{\infty+\varepsilon' i}\frac{e^{iz}-e^{-iz}}{2iz}dz = \int_{-\infty}^{\infty}\frac{\sin z}{z}dz = \pi$$

$a>0$のとき

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ipx}\frac{\sin(ap)}{p}dp = \left.\begin{cases} \frac{1}{2} & (-a<x<a) \\ 0 & (x<-a \text{ or } a<x) \end{cases}\right\} = \chi_{[-a,a]}(x).$$



$\to \int_{-\infty}^{\infty}\frac{\sin x}{x}dx$

$$\boxed{u_{tt} = u_{xx} - u, \quad u(0,x) = a(x), \quad u_t(0,x) = b(x)}$$

$$\hat{u}_{tt} = -(p^2+1)\,\hat{u}, \quad \hat{u}(0,p) = \hat{a}(p), \quad \hat{u}_t(0,p) = \hat{b}(p)$$

$$\hat{u}(t,p) = \hat{a}(p)\cos\left((p^2+1)^{\frac{1}{2}} t\right) + \hat{b}(p)\,\frac{\sin\left((p^2+1)^{\frac{1}{2}} t\right)}{(p^2+1)^{1/2}}$$